Program : **Diploma in Information Technology**	
Course Code : **3268**	Course Title: **SQL and No SQL Lab**
Semester : **3**	Credits: **2.5**
Course Category: **Program Core**	
Periods per week: **4 (L:1 T:0 P:3)**	Periods per semester: **60**

Course Objectives:

- To provide practical knowledge on basic operations in MySQL and MongoDB databases.
- To provide practical knowledge on query processing using advanced MySQL queries.
- To provide practical knowledge on database problems using MySQL programming techniques.

Course Pre-requisites:

Topic	Course Code	Course Name	Semester
Knowledge of basics of Computer		Introduction to IT systems	1
Basic Knowledge of IT skills and basic problem solving		Problem solving and Programming	2

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Implement basic commands in MySQL	12	Applying
CO2	Implement Aggregate functions and Special clauses MySQL	11	Applying
CO3	Develop Stored Procedures and Stored Functions in MySQL	18	Applying
CO4	Implement basic database operations in MongoDB.	16	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	Implement basic commands in MySQL		
M1.01	Implement DDL commands and integrity constraints in MySQL.	4	Applying
M1.02	Implement DML commands in MySQL	4	Applying
M1.03	Implement DCL commands in MySQL.	4	Applying
CO2	Implement Aggregate functions and Special clauses MySQL		
M2.01	Implement Aggregate functions in MySQL.	5	Applying
M2.02	Implement special clauses in MySQL.	6	Applying
	Lab Exam – I	1.5	
CO3	Develop Stored Procedures and Stored Functions in MySQL		
M3.1	Develop programs using Stored Procedures in MySQL	5	Applying
M3.2	Develop programs using Stored Functions in MySQL	5	Applying
M3.3	Develop programs using Stored Procedure and Cursor	8	Applying
CO4	Implement basic database operations in MongoDB		
M4.1	Install and configure MongoDB.	1	Applying
M4.2	Implement create operation in MongoDB.	3	Applying
M4.3	Implement read operation in MongoDB.	4	Applying
M4.4	Implement update operation in MongoDB.	4	Applying
	Open Ended Project	4	

	Lab Exam – II	1.5	

**** **Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation). Students can do open-ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation and a score of 15 marks. Students should prepare a separate report on open ended experiment of their choice.

Example: Develop programs such as

- SALARY BILL OF AN EMPLOYEE,
- STUDENT MARKLIST PREPARATION,
- CUSTOMER BILLING SYSTEM using stored procedure and cursor.
- Perform CURD operations in a MongoDB database.

Text / Reference

T/R	Book Title/Author
T1	**PHP and MySQL web development**-Fifth edition-Luke Welling, Laura Thomson
T2	Sam’s teach6 Yourself NoSQL with6 Mong5oDB- Brad Dayley

Online Resources

Sl.No	Website Link
1	https://www.tutorialspoint.com/mysql/index.htm
2	https://www.tutorialspoint.com/mongodb/index.htm